Incoming preparation constraints and quadratic source budgets for scalar shells

Ricardo Maldonado
Independent Researcher

4 November 2026

Working manuscript for author and specialist review

Abstract

We distinguish a prescribed present perturbation from an incoming physical preparation for a five-dimensional Einstein–scalar shell. The registered compact master packet has nonzero velocity and both principal characteristic channels; their normalized imbalance is at most $\ell^2/4$ for $\ell = 10^{-6}$. This diagnostic neither identifies a physical energy flux nor excludes an incoming past state that subsequently scatters. For the actual first canonical jet, a tilted five-channel energy gives a conditional continuous source-to-shell estimate with growth rate 301/100. Integrating retained quadratic source bounds over five clipped cells of total area 9/32 yields an exact source ledger. Its zero-initial-data trace bound is auxiliary, weaker than a retained direct estimate, and does not supply a constraint-compatible physical second state. On a separately declared past affine null cut, the Raychaudhuri constraint determines an explicit nonpositive second expansion and logarithmic area response after free second scalar data and corner constants are fixed. Actual second area has an additional nonnegative first-variation square. A restricted inverse estimate also identifies admissible present preparations from a normalized physical scalar history. None of these statements selects a past state, a preparation radius, a matter-production mechanism, or a cosmological origin.

1 Preparation question and retained model

An initial perturbation on a supplied Cauchy slice does not specify how that state was prepared. This distinction is especially relevant at second order: the first perturbation fixes quadratic bulk and junction sources, while the second constraints still permit additional homogeneous data. Previous preparation comparisons for this model control shell-volume differences on a restricted canonical fiber by a geometric initial profile and its H^1 norm [7, 8]. Here we ask what an incoming prescription would have to add. The results are a present-packet diagnostic, an actual-source ledger, one solved null constraint, and an inverse estimate on an admissible present-state image. They are not a construction of the full past problem.

We retain the action normalization

$$S_{\mathrm{bulk}} = \int_{\mathcal{M}} \left(\frac{R}{2} - \frac{|\nabla\phi|^2}{2} - V(\phi) \right) \mathrm{d}\mu_g, \qquad \mathrm{Ric}_{AB} = \partial_A\phi\,\partial_B\phi + \frac{2}{3} V g_{AB}, \quad \Box_g\phi = V'(\phi). \tag{1}$$

Two identical interiors meet at a pure-tension shell. In the retained outward-normal convention their one-copy laws are

$$K^{\mathrm{out}}_{\mu\nu} = \frac{\sigma}{6}\gamma_{\mu\nu}, \qquad n_{\mathrm{out}}\phi = -\frac{\sigma'}{2}. \tag{2}$$

The fixed model uses

$$\begin{aligned} W(\phi) &= 1-\phi+\frac{\phi^3}{3}, & V &= \frac{(W')^2}{2}-\frac{2W^2}{3},\\ \sigma(\phi) &= 2W(\phi)+10^{-3}(1+c_\sigma\phi), & c_\sigma &= \frac{96714321425692}{161839258930731}. \end{aligned} \tag{3}$$

Let κ be the unit three-sphere metric. The retained spherical background has

$$g_0 = \mathrm{d}y^2+\rho(y)^2[-\mathrm{d}t^2+\cosh^2 t\,\kappa] = \rho(z)^2[-\mathrm{d}t^2+\mathrm{d}z^2+\cosh^2 t\,\kappa], \quad \mathrm{d}y=-\rho\,\mathrm{d}z. \tag{4}$$

The shell is $z=0$, $\rho_b=\rho(0)$, $a=\rho_y/\rho$, and $p=\phi_{0,y}\neq 0$ on the collar in use. Set

$$\ell=10^{-6}, \qquad x=t/\ell, \qquad r=z/\ell, \qquad T=\frac{5}{4}, \qquad \tau_*=\rho_b\ell T. \tag{5}$$

All estimates below use this fixed model normalization; no measured SI units are assigned. A subscript j denotes the actual derivative $\partial_\varepsilon^j|_{\varepsilon=0}$, so a second perturbation enters a Taylor series as $\varepsilon^2 X_2/2$.

Retained premise 1. *The background, the common first packet, the retained analytic reductions, the regular exact-family and uniqueness premises, and the continuous future-cone operator caps stated in Section 4 are shared inputs. The present calculation does not reprove them. In particular, finite source certificates do not by themselves prove a coefficient bound at every point.*

The physical slice, shell anchor, scalar-level component, and background scalar labels used in the earlier geometric preparation class remain supplied structure. A new past prescription must establish a compatible development before it can be tested against that class.

2 The actual first packet and its two principal channels

The registered master data are

$$\begin{aligned} w(0,z) &= q_\ell(z), & w_t(0,z) &= e^{-z}q_\ell(z),\\ q_\ell(z) &= \begin{cases} \exp\left(4-\dfrac{\ell^2}{(z-\ell)(2\ell-z)}\right), & \ell<z<2\ell,\\ 0, & \text{otherwise.}\end{cases} \end{aligned} \tag{6}$$

The function is smooth and flat at both endpoints. In scaled variables $W_0=Q(r)=q_\ell(\ell r)$ and $W_{x,0}=\ell e^{-\ell r}Q$. At $r=3/2$, $Q=1$ and $Q_r=0$, so both W_x+W_r and W_x-W_r are strictly positive. Background time symmetry, $K_0=\Pi_0=0$, does not make this perturbation time symmetric.

The first physical metric and scalar data are reconstructed together. Writing $\mathcal{Y}=n_1^2-n_2^2$ on the unit three-sphere, define

$$\begin{aligned} \alpha &= (\rho/\rho_b)^{3/2}p/p_b, & C &= \alpha/\rho^2, & B &= 3\alpha/(\rho^3 p),\\ k_{\mathrm{bg}} &= \rho(5a/2-V'/p), & u &= Cq_\ell, & v &= B(q_{\ell,z}+k_{\mathrm{bg}}q_\ell). \end{aligned} \tag{7}$$

Their actual velocities, not freely assigned zero velocities, are

$$u_t=e^{-z}u, \qquad v_t=e^{-z}\left(v-\frac{3u}{\rho p}\right)=Be^{-z}[q_{\ell,z}+(k_{\mathrm{bg}}-1)q_\ell]. \tag{8}$$

In radial/angular orthonormal order the Cauchy data are

$$\begin{aligned} \gamma_1 &= \mathrm{diag}(-4u,2u,2u,2u)\mathcal{Y},\\ K_1 &= \rho^{-1}\,\mathrm{diag}(-2u_t,u_t,u_t,u_t)\mathcal{Y}, \quad \phi_1=v\mathcal{Y}, \quad \Pi_1=v_t\mathcal{Y}/\rho. \end{aligned} \tag{9}$$

For the convention $\pi^{ij} = \sqrt{\gamma}(K^{ij} - K\gamma^{ij})/2$, these imply

$$\frac{\pi_1}{\sqrt{\gamma_0}} = \operatorname{diag}\Big(-\frac{3u_t}{2\rho}, 0, 0, 0\Big)\,\mathcal{Y}, \qquad \frac{\varpi_1}{\sqrt{\gamma_0}} = \Pi_1. \tag{10}$$

Harmonic completion further fixes

$$\begin{aligned} &N_1 = \rho u\mathcal{Y}, && \beta_1 = 0, && \dot N_1 = \rho u_t\mathcal{Y}, \\ &\dot\beta_1^y = -6\rho^2(u_y + 4au)\mathcal{Y}, && \dot\beta_1^a = 0 \quad (a = 1,2,3), && \dot\gamma_1 = 2\rho K_1, \quad \dot\phi_1 = v_t\mathcal{Y}. \end{aligned} \tag{11}$$

The quadratic sources used below are joined to this complete canonical and harmonic one-jet. The retained 31 October velocity correction concerned a *second-response approximation*; it did not replace (6) by time-symmetric first data.

For a transparent principal-part diagnostic, introduce

$$E_\pm = \frac{1}{4}\int_\ell^{2\ell} (e^{-z}q_\ell \pm q_{\ell,z})^2\,\mathrm{d}z. \tag{12}$$

These positive quadratic weights refer to the normalized master, not to a physical Noether energy or a measured incoming flux. Integration by parts using flat endpoints gives

$$\begin{aligned} E_+ - E_- &= \int e^{-z}q_\ell q_{\ell,z}\,\mathrm{d}z = \frac{1}{2}\int e^{-z}q_\ell^2\,\mathrm{d}z > 0, \\ E_+ + E_- &= \frac{1}{2}\int (q_{\ell,z}^2 + e^{-2z}q_\ell^2)\,\mathrm{d}z, \\ \delta := \frac{E_+ - E_-}{E_+ + E_-} &= \frac{\int e^{-z}q_\ell^2\,\mathrm{d}z}{\int (q_{\ell,z}^2 + e^{-2z}q_\ell^2)\,\mathrm{d}z} \le \frac{\ell^2}{4} = \frac{1}{4000000000000}. \end{aligned} \tag{13}$$

Indeed, endpoint Cauchy–Schwarz gives $|f(z)|^2 \le \min(z-\ell, 2\ell - z)\int |f'|^2$ for a function vanishing at both endpoints. Its integral bounds $\int f^2$ by $\ell^2\int (f')^2/4$. Since $e^{-z} \le 1$, this proves the last line without new bump evaluations. The normalized weights are $(1 \pm \delta)/2$: each differs from $1/2$ by at most $1/8000000000000$.

For the free principal part, $W_x + W_r$ propagates toward the shell, whereas $W_x - W_r$ propagates away. Pure propagation toward the shell now would require $Q_r = \ell e^{-\ell r}Q$; its zero endpoint value forces the zero solution, contrary to (6). The opposite one-way condition also fails. The actual equation includes a potential and a reflecting, coupled shell. Thus these present data may in principle result from an incoming past state after scattering. The calculation establishes neither that history nor its impossibility.

3 Forced mean equations and the physical corner

Use the retained five mean fields $U = (K, L, N, J, w)$ with $H = \operatorname{diag}(1, 2, 1, 3, 1)$. Let P_D project onto L and $P_N = 1 - P_D$; these projections are orthogonal in H. The actual second mean equations on the future cone are

$$\begin{aligned} U_{xx} - U_{rr} &= AU_x + BU_r + D_{\mathrm{op}}U + S_0, \\ P_D U = 0, \qquad P_N U_r &= Mu + G_0, \qquad u = P_N U(x, 0). \end{aligned} \tag{14}$$

The boundary derivative in (14) points toward *increasing* r, away from the shell. In terms of the quadratic reduced Einstein, scalar, and one-copy junction Hessians, the adapters are

$$\begin{aligned} S_0 &= P_0(-2\ell^2\rho^2 Q_{E,\mathrm{red}},\ \ell^2\rho^2 Q_\phi), \\ G_0 &= \ell\rho_b(-G_{N,0},\ 0,\ 2Q_{R00,0},\ 2Q_{RJ,0},\ Q_{P,0}), \\ G_{N,0} &= 2Q_{R00,0} - 6Q_{RJ,0} - 2Q_{Cy,0}. \end{aligned} \tag{15}$$

Here P_0 is angular mean projection, $Q_{RJ,0}$ is the mean angular junction row, and the subscript 0 on a source row denotes that projection, not a vanishing source. The sign uses $\partial_r = -\ell\rho\partial_y$. The source is a derivative forcing in the Robin law, not its time integral. There is no additional factor $1/2$ in these actual-second-variation adapters.

For example, in the geometric shell Hessian map write first normal lapse $H_f = A_f\mathcal{Y}$, induced components $m_{00} = D_f\mathcal{Y}$ and $m_{ab} = L_f\mathcal{Y}\kappa_{ab} + M_f\nabla_a\nabla_b\mathcal{Y}$, and scalar $\eta = s_f\mathcal{Y}$. With

$$\begin{aligned} c_f &= aA_f^2/4 - \sigma' A_f s_f/6 - \sigma'' s_f^2/6, \\ d_f &= aA_f + \sigma' s_f/3, \qquad q_f = pA_f^2/4 + \sigma'' A_f s_f/2 + 2s_f^2, \end{aligned} \tag{16}$$

the retained angular moments give

$$Q_{R00,0} = \frac{-c_f - d_f D_f}{6}, \quad Q_{RJ,0} = \frac{c_f - d_f L_f}{6} + \frac{4d_f M_f}{9}, \quad Q_{P,0} = \frac{q_f}{6}. \tag{17}$$

Together with the Codazzi row, this records which geometric source is being bounded; it does not substitute a test scalar for the actual metric jet.

Set $J_\pm = U_x \pm U_r$. The characteristic and shell identities are exactly

$$\begin{aligned} (\partial_x \mp \partial_r)J_\pm &= AU_x + BU_r + D_{\mathrm{op}}U + S_0, \\ P_N J_- &= P_N J_+ - 2(Mu + G_0), \\ u_x &= P_N J_+ - Mu - G_0, \qquad P_D J_- = -P_D J_+. \end{aligned} \tag{18}$$

Thus a shell incoming characteristic combination cannot be prescribed independently while keeping the junction condition fixed. At a regular Cauchy–shell corner, with $U_0 = U(0,\cdot)$ and $V_0 = U_x(0,\cdot)$, one must have

$$\begin{aligned} P_D U_0 = P_D V_0 = 0, \qquad P_N U_{0,r} &= M(0)u_0 + G_0(0), \\ P_N V_{0,r} &= M_x(0)u_0 + M(0)P_N V_0 + G_{0,x}(0). \end{aligned} \tag{19}$$

Higher regularity imposes further differentiated identities. The actual packet has an open background collar at the initial shell: the needed source and data jets vanish there. This verifies these corner rows for the retained packet. It does not verify a corner for a newly chosen past state.

4 A continuous tilted energy with actual forcing

On $0 \le r \le T - x$ put $V = U_x$, $P = U_r$, $k = \alpha = 1/2$, and

$$e = \frac{|V|_H^2 + |P|_H^2 + 2k\langle V, P\rangle_H + |U|_H^2}{2}, \qquad E(x) = \int_0^{T-x} e(x,r)\,\mathrm{d}r + \frac{\alpha}{2}|u(x)|_H^2. \tag{20}$$

The cross term is a standard tilted wave energy device [6]. It is positive since $|k| < 1$; it is not identified here with a physical Noether energy or a preparation law. The retained continuous operator bounds, in the H-induced norm, are

$$\|A\|, \|B\| \le 10^{-4}, \qquad \|D_{\mathrm{op}}\| \le 10^{-6}, \qquad \|M\| \le \mu := \frac{1}{2000}. \tag{21}$$

They are conditional analytic premises on the existing future cone. They are not new all-point proofs, and give no cap on a new past domain or for a different first state.

Proposition 1. *Under* (21)*, every regular solution of* (14) *satisfies, including the zero-energy case,*

$$E'(x) \le \frac{301}{100}E(x) + \frac{1}{2}\|S_0(x,\cdot)\|_{L^2(H)}^2 + \frac{5002001}{4000000}|G_0(x)|_H^2. \tag{22}$$

No selfadjointness, positivity, or time independence of M is required.

Proof. With $F = AV + BP + D_{\mathrm{op}}U$, direct differentiation gives $e_x = f_r + \langle V + kP, F + S_0\rangle_H + \langle U, V\rangle_H$, where $f = \langle V, P\rangle_H + k(|V|_H^2 + |P|_H^2)/2$. For $a_* = b_* = 10^{-4}$ and $d_* = 10^{-6}$, the homogeneous bulk production is at most $c_V|V|_H^2 + c_P|P|_H^2 + c_U|U|_H^2$, with

$$\begin{aligned} c_V &= a_* + (b_* + ka_*)/2 + d_*/2 + 1/2, \\ c_P &= kb_* + (b_* + ka_*)/2 + kd_*/2, \qquad c_U = d_*/2 + kd_*/2 + 1/2. \end{aligned}$$

Its rate is bounded by $\max\{2c_V/(1-k), 2c_P/(1-k), 2c_U\} = 2.000702$. At the homogeneous Robin boundary, put $j = P_N(V + P)$ and $\gamma_* = \alpha + (1-k)\mu$. Completing one square gives

$$-f_N + \alpha\langle u, P_N V\rangle_H \le -\frac{k}{4}|j|_H^2 + \left[\frac{\gamma_*^2}{k} + (1-k)\mu^2 + \alpha\mu\right]|u|_H^2.$$

The coefficient relative to the boundary energy is 2.003001. This leaves $-|j|_H^2/8$ available for absorption. The moving outer flux is

$$-\frac{1-k}{2}|V - P|_H^2 - \frac{1}{2}|U|_H^2 \le 0,$$

and the physical Dirichlet contribution is nonpositive. These conclusions follow by direct integration by parts for this mixed boundary problem; they are not an application of a theorem assuming a different boundary law.

For the bulk forcing one has the exact square identity

$$2e - |V + kP|_H^2 = (1 - k^2)|P|_H^2 + |U|_H^2 \ge 0.$$

Consequently $\langle V + kP, S_0\rangle_H \le e + |S_0|_H^2/2$, adding at most one to the bulk rate. Expanding the Robin boundary expression about its homogeneous value adds exactly

$$-\tfrac{1}{2}\langle j, G_0\rangle_H + \langle Mu, G_0\rangle_H + \tfrac{1}{2}|G_0|_H^2 - \tfrac{1}{2}\langle u, G_0\rangle_H.$$

Use the reserved negative square and then Young's inequality:

$$\begin{aligned} -|j|_H^2/8 + |j|_H|G_0|_H/2 &\le |G_0|_H^2/2, \\ (\mu + 1/2)|u|_H|G_0|_H &\le |u|_H^2/4 + (\mu + 1/2)^2|G_0|_H^2. \end{aligned}$$

The first term in the second line is exactly the boundary summand in E. Thus the total boundary source coefficient is $1 + (\mu + 1/2)^2 = 5002001/4000000$. Both enlarged homogeneous rates are below $301/100$, proving (22). □

With $C = 301/100$ and $C_G = 5002001/4000000$, integration gives

$$\begin{aligned} E(x) \le e^{Cx}E(0) + \int_0^x e^{C(x-s)}\left[\frac{1}{2}\|S_0(s,\cdot)\|_{L^2(H)}^2 + C_G|G_0(s)|_H^2\right]\mathrm{d}s, \\ e^{CT} = e^{301/80} < 44, \qquad |w(x,0)|^2 \le |u(x)|_H^2 \le 4E(x). \end{aligned} \tag{23}$$

The exponential upper bound follows from a rational Taylor sum and a geometrically bounded tail. The estimate retains the coupled metric, scalar, and nonselfadjoint Robin feedback. It does not assume that a single uncoupled scalar energy controls the other fields.

5 The clipped source ledger and its interpretation

The retained actual-jet certificates give global source bounds

$$F_H = 17387513156573, \qquad G_H = 3851400. \tag{24}$$

The future cone is $x \geq 0$, $r \geq 0$, $x + r \leq T$, of area $25/32$. Finite propagation and the open initial collar give the exact zero wedge $x + r \leq 1$. Hence the active bulk has area $9/32$, and the active shell segment is $1 \leq x \leq 5/4$, of length $1/4$. The old half-unit certificate grid intersects the active cone in five nonempty polygons, shown in Figure 1. Integrating each retained cell cap only over its clipped polygon avoids a new source call.

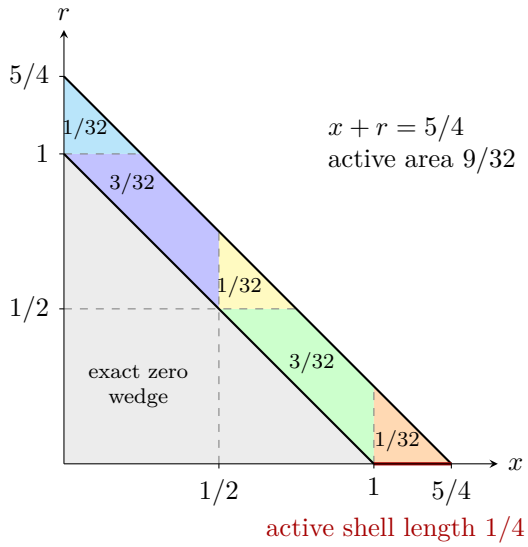


Figure 1: Clipping the retained half-unit cells to $1 < x + r \leq 5/4$. Labels are exact polygon areas; cell boundaries are dashed. The red shell segment carries the active Robin source. This is a source-integration domain, not a new past development.

Let

$$I_{\mathrm{bulk}} = \int_0^T \|S_0(x,\cdot)\|^2_{L^2(H)}\,\mathrm{d}x, \quad I_{\mathrm{bd}} = \int_0^T |G_0(x)|^2_H\,\mathrm{d}x, \quad B_{\mathrm{src}} = \frac{1}{2} I_{\mathrm{bulk}} + C_G I_{\mathrm{bd}}. \tag{25}$$

Exact rational integration of the retained cell caps gives the outward integer bounds

$$\begin{aligned} I_{\mathrm{bulk}} &\leq 852641812927542852015 2526, \\ I_{\mathrm{bd}} &\leq 3708320490000, \\ B_{\mathrm{src}} &\leq 426320906464235151577 6089. \end{aligned} \tag{26}$$

For comparison, using only the active area and the global cap gives $I_{\mathrm{bulk}} \leq 9F_H^2/32 < 8.50291\,10^{25}$; the cell integration gives $I_{\mathrm{bulk}} < 8.52642\,10^{24}$. The source amplitudes were retained rather than newly sampled. The arithmetic authenticates the selected old inputs and integrates their enclosures; it does not reconstruct the quadratic source function.

Equations (23)–(26) imply

$$|w(x,0)|^2 \leq 176\big(E(0) + B_{\mathrm{src}}\big). \tag{27}$$

For the *auxiliary* forced problem with $U_0 = V_0 = 0$, this yields

$$\sup_{0\le x\le T} |w(x,0)| \le 27392057158547. \tag{28}$$

This is not an improved numerical response prediction: the retained direct Duhamel estimate of about $5.4\,10^{12}$ is smaller. The purpose of (28) is a continuous, source-norm-based ledger.

Crucially, the auxiliary zero state is not a physical zero second Einstein–scalar state. The second constraints read $PX_2 = -D^2\mathcal{C}[X_1, X_1]$. For the actual packet the retained Hamiltonian Hessian witness is less than -10^{19} at its center. Setting *all* second Cauchy fields to zero therefore fails that constraint. An affine seed before solving the constraints is a different object. Nor does a bound on w alone omit the common first-quadratic proper-clock, volume-measure, and Gauss terms required for an absolute shell-volume or shell-expansion coefficient.

6 A declared past cut and its forced null constraint

6.1 Affine geometry and background check

One candidate past initial characteristic surface is the *outgoing-relative-to-the-shell* cut

$$\mathcal{C}_- = \{t - z = t_-\}, \qquad t_- < -2\ell, \tag{29}$$

restricted to a segment of the actual positive collar preceding $t = 0$. Its shell corner is $(t_-, 0)$. The strict margin matters for encompassing the present packet. An incoming pulse crossing this outgoing cut could later move toward the shell; the cut orientation is not the pulse direction. A future affine background generator is

$$k_0 = c\rho^{-2}(\partial_t + \partial_z), \qquad c > 0, \qquad R_0(\lambda) = \rho(z(\lambda))\cosh t(\lambda). \tag{30}$$

It increases z away from the shell. The positive constant c must be fixed by a declared corner-clock normalization. Primes in this section mean affine λ derivatives. The background screen has dimension three, and its expansion and scalar derivative are

$$\vartheta_0 = 3R_0'/R_0 = 3c\rho^{-2}(\tanh t - \rho a), \qquad \Phi_0' = -cp/\rho. \tag{31}$$

In particular, the screen radius is not ρ alone.

An exact background check fixes the signs and normalization. The static equations imply $\rho_z = -a\rho^2$, $a_y = -1/\rho^2 - p^2/3$, and hence $a_z = 1/\rho + \rho p^2/3$. Put $A_0 = \rho a$, $T_0 = \tanh t$, and $\partial_+ = \partial_t + \partial_z$. Then

$$\partial_+(T_0 - A_0) = A_0^2 - T_0^2 - \rho^2 p^2/3, \qquad \partial_+(\rho^{-2}) = 2A_0\rho^{-2}.$$

Substituting in (31) cancels every A_0, T_0 term in $\vartheta_0' + \vartheta_0^2/3$, leaving $-c^2p^2/\rho^2 = -(\Phi_0')^2$. This algebraic identity does not constitute a numerical enclosure on a new past domain.

For a smooth family, choose physical null cuts $\mathcal{C}_\varepsilon$ and pull them back to fixed (λ, ω) so that $k_\varepsilon = \partial_\lambda$ is affine for every family member with the specified corner normalization. Both the embedding and the normalization belong to the new state prescription. Let $q_{\varepsilon,AB}$ be the positive screen metric and Φ_ε the pulled-back scalar. Define

$$\mathcal{B}_{AB} = \frac{1}{2}q'_{AB}, \qquad \vartheta = q^{AB}\mathcal{B}_{AB}, \qquad \varsigma_{AB} = \mathcal{B}_{AB} - \frac{\vartheta}{3}q_{AB}. \tag{32}$$

Twist vanishes because the cut is a null hypersurface. At the background $q_0 = R_0^2\kappa$ and $\varsigma_0 = 0$. The coefficients Φ_j include cut and generator changes through the pullback. An old harmonic-coordinate scalar coefficient cannot be substituted without performing that transport.

6.2 Second expansion and logarithmic area

For (1), $\mathrm{Ric}(k,k) = (\Phi')^2$ since the potential term vanishes on a null vector. The affine optical constraint is

$$\vartheta' + \frac{\vartheta^2}{3} + |\varsigma|_q^2 + (\Phi')^2 = 0. \tag{33}$$

Its first and actual second variations are

$$\vartheta_1' + \frac{2\vartheta_0}{3}\vartheta_1 = -2\Phi_0'\Phi_1', \tag{34}$$

$$\vartheta_2' + \frac{2\vartheta_0}{3}\vartheta_2 = -\frac{2}{3}\vartheta_1^2 - 2|\varsigma_1|_{q_0}^2 - 2(\Phi_1')^2 - 2\Phi_0'\Phi_2'. \tag{35}$$

No ς_2 term or extra inverse-metric term survives at this order because $\varsigma_0 = 0$. The factors two follow from actual second variation. The fixed affine pullback is essential: a changing nonaffine generator would introduce additional terms.

The integrating factor in (35) is R_0^2. Thus generally

$$\begin{aligned} R_0(\lambda)^2\vartheta_2(\lambda) = {} & R_0(\lambda_0)^2\vartheta_2(\lambda_0) \\ & - 2\int_{\lambda_0}^{\lambda} R_0(s)^2 \left[\frac{\vartheta_1^2}{3} + |\varsigma_1|^2 + (\Phi_1')^2 + \Phi_0'\Phi_2'\right](s)\,\mathrm{d}s. \end{aligned} \tag{36}$$

The mixed scalar term has no fixed sign. Now *declare* the candidate free-state and corner conditions

$$\Phi_2 = 0 \quad \text{on } \mathcal{C}_-, \qquad \vartheta_2(\lambda_0,\omega) = 0. \tag{37}$$

These are new assumptions, not consequences of the action or equivalent to zero scalar second data on the present Cauchy slice. Define $S_1 = \vartheta_1^2/3 + |\varsigma_1|^2 + (\Phi_1')^2 \geq 0$. Then the unique response of this constraint is

$$\vartheta_2(\lambda,\omega) = -2R_0(\lambda)^{-2}\int_{\lambda_0}^{\lambda} R_0(s)^2 S_1(s,\omega)\,\mathrm{d}s \leq 0. \tag{38}$$

It is strictly negative downstream of a positive integral. The first expansion is itself constrained by (34); it is not assignable independently of the first scalar data.

Write $\mathcal{L}_\varepsilon = \log(\,\mathrm{d}\mu_{q_\varepsilon}/\,\mathrm{d}\mu_{q_0})$. Since $\mathcal{L}_j' = \vartheta_j$, the additional corner choice $\mathcal{L}_2(\lambda_0) = 0$ gives

$$\mathcal{L}_2(\lambda,\omega) = -2\int_{\lambda_0}^{\lambda} R_0(s)^2 S_1(s,\omega)\left[\int_s^{\lambda} R_0(v)^{-2}\,\mathrm{d}v\right]\mathrm{d}s \leq 0. \tag{39}$$

Actual area is different. At a fixed affine section,

$$\frac{A_2(\lambda)}{A_0(\lambda)} = \left\langle \mathcal{L}_2 + \mathcal{L}_1^2 \right\rangle_\kappa. \tag{40}$$

The nonnegative square prevents an actual-area sign conclusion from (39) alone. Moreover ϑ_2 is not the proper-time shell expansion Θ_2, and these null sections are not the scalar-level sections on a physical Cauchy slice used in the geometric profile D. Relating them requires clock, embedding, scalar-label, and canonical transport.

6.3 Extent of the construction

Equations (36)–(39) solve one necessary null Einstein constraint once free data and corner constants are supplied. They do not solve the remaining null constraints, the scalar wave equation, the shell junction law, or a nonlinear characteristic initial/boundary problem. The present mixed packet does not already provide its first Φ_1, ς_1 on this past cut. Obtaining them requires a past development or an independently selected incoming state proven to yield that packet.

This separation is standard in characteristic Einstein–scalar work. Holzegel and Warnick, for example, specify characteristic data, constraint ordinary differential equations, boundary data, and a well-posedness theorem for a different spherical four-dimensional AdS problem [3]. Their theorem is not a theorem for this five-dimensional moving-shell problem. The scalar-brane equations of Langlois and Rodríguez-Martínez also exhibit wave mixing and junction restrictions [2]; they do not determine our incoming state.

7 Shell exchange does not supply a matter-production state

The physical meaning of the current shell action is restricted. For $S_{\mu\nu} = -\sigma(\phi)\gamma_{\mu\nu}$, its rest-frame energy density and pressure are

$$\rho_\sigma = \sigma, \qquad P_\sigma = -\sigma. \tag{41}$$

The two-copy Codazzi balance and (2) give

$$D^\mu S_{\mu\nu} = 2n_{\mathrm{out}}\phi\, D_\nu\phi = -\sigma' D_\nu\phi, \qquad \dot\rho_\sigma + \Theta(\rho_\sigma + P_\sigma) = \sigma'\dot\phi. \tag{42}$$

For a compact shell section transported by the stated normal proper-time flow, this is equivalently

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\int \sigma\, \mathrm{d}\mathcal{V} - \int \sigma\Theta\, \mathrm{d}\mathcal{V} = \int \sigma'\dot\phi\, \mathrm{d}\mathcal{V}. \tag{43}$$

Exchange in the pure-tension sector is therefore not a radiation fluid. The projected bulk Weyl term in effective brane equations is likewise not by itself a thermal matter component [4].

A new local shell matter action $S_m = \int L_m(\chi, \phi, \gamma)\, \mathrm{d}\mu_\gamma$, with worldvolume measure $\mathrm{d}\mu_\gamma = \sqrt{-\gamma}\, \mathrm{d}^4x$, would add

$$\tau_{\mu\nu} = -\frac{2}{\sqrt{-\gamma}}\frac{\delta S_m}{\delta\gamma^{\mu\nu}}, \qquad J_m = \frac{1}{\sqrt{-\gamma}}\frac{\delta S_m}{\delta\phi}.$$

For a coupling algebraic in ϕ and with no scalar-derivative or extrinsic-curvature terms, $J_m = \partial_\phi L_m$. In our conventions,

$$\begin{aligned} S_{\mu\nu} &= -\sigma\gamma_{\mu\nu} + \tau_{\mu\nu}, \qquad K^{\mathrm{out}}_{\mu\nu} = \tfrac{1}{2}(S_{\mu\nu} - S\gamma_{\mu\nu}/3), \\ 2n_{\mathrm{out}}\phi &= -\sigma' + J_m, \qquad D^\mu\tau_{\mu\nu} = J_m D_\nu\phi. \end{aligned} \tag{44}$$

The sign of a conformal source must follow its explicit action. For $S_m[\chi, A(\phi)^2\gamma]$ and the stress convention in (44), variation gives $J_m = (\,\mathrm{d}\log A/\,\mathrm{d}\phi)\tau^\mu{}_\mu$. An inverse conformal metric $\Omega^{-2}\gamma$ gives the corresponding minus sign. Thus a classically traceless radiation sector has no scalar exchange from this conformal coupling alone. This observation does not exclude other couplings, massive matter, or quantum effects.

Maeda and Wands distinguish scalar-brane exchange from the undetermined bulk information [1]; Langlois and Rodríguez-Martínez discuss the corresponding coupling and trace restrictions [2]. Reheating mechanisms, such as the explicit interacting fields and state evolution studied by Kofman, Linde, and Starobinsky [5], require those ingredients. They are absent from (1)–(3) as an additional shell matter sector. We assert neither classical radiation production nor quantum particle creation from the present geometric perturbation.

8 What an incoming prescription must still determine

The source ledger applies to the unchanged first packet on its certified *future* domain. It cannot be extended to an uncomputed past source history by relabeling time. The null sign result depends on declared free second state data; it supplies neither an H^1 radius nor a complete incoming canonical state. A physical retarded prescription must specify the past clock and domain, nontrivial incoming data or a generating process, all constraint-compatible free data, and the physical shell corner. With background past data, no incoming perturbation, and no added source, uniqueness cannot create the fixed nonzero packet.

Even a completed past solution need not arrive in the retained compact h-only canonical fiber with every other second sector fixed. That membership must be checked before its scalar-level preparation profile can be inserted into the prior H^1 response theorem [8]. The present results neither enlarge that theorem to arbitrary canonical data nor select its numerical budget. They give no effective common nonlinear ε interval, and no claim of a physical origin. The energy method, null focusing identity, and conservation identities are standard; the contribution here is their explicit application to the registered packet, source adapters, and restricted preparation map.

A A restricted inverse from physical scalar history

There is a present-data identifiability statement that can be proved without completing the past problem. Fix the background, common first packet, other second canonical sectors, harmonic conditioning, physical shell and anchor of the geometric comparison. The smooth compact metric increment is parameterized by $h = \rho^{-2}\chi_y$ with $\int \rho^2 h \, \mathrm{d}y = 0$ on the packet and flat endpoints; the other added raw canonical sectors vanish. For the continued background three-sphere component of $\{\phi_\varepsilon = \varphi\}$ inside the supplied physical Cauchy slice, let $D = \Delta(\partial_\varepsilon^2 A_\varepsilon)/A_0$ at fixed background scalar label $\varphi = \phi_0(y)$. Let $w_{\mathrm{in}} = -p\zeta$ be the initial conditioned scalar mean *difference*. It is distinct from the evolved scalar channel $w(x, r)$. The geometric scalar-level area profile obeys $D = 3h/2 - 3aw_{\mathrm{in}}/p$. Substitution into the retained constraint reconstruction, with $Z = \chi/(\ell\rho^3)$ and $b = \ell\rho a$, gives

$$Z_r = 3bZ - h, \qquad h_r = 3\ell^2 Z - bh + c_w w_{\mathrm{in}}, \qquad c_w = 2\ell\rho p/3, \qquad Z(1) = h(1) = 0. \tag{45}$$

Thus w_{in} determines h, Z causally on $I = [1, 5/4]$ in this fiber. Whole-packet flatness and moment conditions still restrict its admissible image; arbitrary profiles on I do not automatically satisfy them.

For $Y(f) = \max(\sup_I |f|, \sup_I |f'|)$, retained background bounds give

$$\begin{aligned} b_M &= \frac{917}{62500000}, & c_M &= \frac{131}{10500000000}, \quad H_w = \frac{c_M}{3}, \\ H_{rw} &= (3\ell^2 + b_M)H_w + c_M, & V_w &= 3\ell^2 H_w + c_M. \end{aligned} \tag{46}$$

The two-row matrix norm in (45) is at most $20001/20000$; its exponential on length $1/4$ is below $4/3$. Consequently $|h|, |Z| \le H_w Y(w_{\mathrm{in}})$, $|h_r| \le H_{rw} Y(w_{\mathrm{in}})$. The initial five means are

$$U_0 = (h, 0, 0, h, w_{\mathrm{in}}), \quad U_{0,x} = (0, V_L, 0, 0, 0), \quad U_{0,r} = (h_r, 0, 0, h_r, w_{\mathrm{in},r}), \quad V_L = 3\ell^2 Z + c_w w_{\mathrm{in}}. \tag{47}$$

Define $D_0 = \sup_I |U_0|_H$, $D_r = \sup_I |U_{0,r}|_H$, $D_v = \sup_I |U_{0,x}|_H$, and $N = \max\{(D_0 + TD_v)/3, D_r + D_v\}$. The retained initial transport norm therefore satisfies the non-strict guard, also when $Y = 0$,

$$N \le N_g Y(w_{\mathrm{in}}), \qquad N_g = \frac{20000001}{20000000}. \tag{48}$$

Indeed its direct coefficient is the maximum of $(1 + 2H_w + 2TV_w)/3$ and $1 + 2H_{rw} + 2V_w$, both below N_g.

Let W now denote the full homogeneous five-mean *difference*, and B_{free} its reflected free wave. The retained coupled resolvent estimate, in $X = \max(\sup|U|_H/3, \sup|U_x|_H, \sup|U_r|_H)$, is

$$X(W - B_{\mathrm{free}}) \le \frac{qN}{1-q}, \qquad q = \frac{1201}{3200000}. \tag{49}$$

The initial scalar velocity is zero and its reflection is even, so $(B_{\mathrm{free}})_w(x,0) = w_{\mathrm{in}}(x)$ exactly. Both solutions vanish for $x + r \le 1$. Hence for $m(x) = W_w(x,0)$ and $e = m - w_{\mathrm{in}}$,

$$|e_x| \le \frac{qN}{1-q}, \quad |e(x)| \le (x-1)\frac{qN}{1-q}, \quad Y(e) \le \kappa_* Y(w_{\mathrm{in}}), \quad \kappa_* = \frac{24020001201}{63975980000000} < \frac{47}{125000} < 1. \tag{50}$$

All internal metric and Robin couplings remain in this estimate.

The history has a physical difference interpretation. On an anchored physical shell section at fixed proper clock $\tau = \rho_b \ell x$, let

$$\mathcal{M}_\varepsilon(\tau) = \frac{\int \phi_\varepsilon \, \mathrm{d}\mathcal{V}_\varepsilon}{\int \mathrm{d}\mathcal{V}_\varepsilon}. \tag{51}$$

For two exact families sharing the first perturbation, the difference of their actual second derivatives of (51) is m. The background scalar is constant on the shell, so its term $\phi_b \Delta\mathcal{V}_2$ cancels between numerator and denominator. Common first scalar, measure, and clock quadratic terms cancel in the family difference. A pure second clock shift multiplies the zero derivative of ϕ_b. The fixed shell representative includes the embedding pullback and has zero pure second normal gauge component at the boundary. An unnormalized scalar integral would contain $\phi_b \Delta\mathcal{V}_2$ and is not the quantity in (51).

The triangle inequality yields

$$(1-\kappa_*)Y(w_{\mathrm{in}}) \le Y(m) \le (1+\kappa_*)Y(w_{\mathrm{in}}), \qquad Y(w_{\mathrm{in}}) \le \frac{Y(m)}{1-\kappa_*}. \tag{52}$$

Here the history norm uses x and the initial norm uses r, fixed by the same ℓ; they are not SI norms. Applying (52) to two admissible profiles proves causal injectivity on the image. Equal complete histories on I force equal w_{in}, then equal h, Z, ζ there. No exterior state, other canonical sector, or higher nonlinear coefficient is determined. There is no onto theorem for arbitrary C^1 histories and no assertion that such a history has been empirically observed.

Finally, setting a present homogeneous scalar contribution to zero selects a base only relative to a chosen particular/homogeneous decomposition. Changing that arbitrary split changes the prescription. It is not a retarded physical law without independently supplied past data.

B Reproducibility and scope of verification

The accompanying capsule retains the source certificates and analytic premises separately from the new exact postprocessing. Its standard-library replay authenticates the selected retained inputs, clips five old cells, and verifies rational energy and inverse guards without new background, bump, source-oracle, or PDE calls. A separate displayed-bounds audit checks the printed numbers against the canonical report. The editable source, two fresh deterministic three-pass PDF builds, and rendering and independent mathematical review receipts accompany this manuscript. These checks do not reprove the retained analytic premises or solve the missing past problem.

References

[1] K. Maeda and D. Wands, *Dilaton-gravity on the brane*, Phys. Rev. D **62**, 124009 (2000), arXiv:hep-th/0008188.

[2] D. Langlois and M. Rodríguez-Martínez, *Brane cosmology with a bulk scalar field*, Phys. Rev. D **64**, 123507 (2001), arXiv:hep-th/0106245.

[3] G. H. Holzegel and C. M. Warnick, *The Einstein–Klein–Gordon–AdS system for general boundary conditions*, arXiv:1312.5332.

[4] T. Shiromizu, K. Maeda, and M. Sasaki, *The Einstein equations on the 3-brane world*, Phys. Rev. D **62**, 024012 (2000), arXiv:gr-qc/9910076.

[5] L. Kofman, A. Linde, and A. A. Starobinsky, *Towards the theory of reheating after inflation*, Phys. Rev. D **56**, 3258–3295 (1997), arXiv:hep-ph/9704452.

[6] H.-O. Kreiss, O. Reula, O. Sarbach, and J. Winicour, *Well-posed initial-boundary value problem for the harmonic Einstein equations using energy estimates*, Class. Quantum Grav. **24**, 5973–5984 (2007), arXiv:0707.4188.

[7] R. Maldonado, *Scalar-level initial geometry and robust shell-volume perturbations*, working manuscript (4 November 2026), dated research capsule.

[8] R. Maldonado, *Integrated geometric control of shell-volume response*, working manuscript (4 November 2026), dated research capsule.